

MOSKVA VNUKOVO, Russian Federation

WMO: 275185

Lat: 55.592N

Lon: 37.261E

Elev: 209

StdP: 98.84

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-19.9	-26.0	0.4	-22.5	-22.8	0.5	-19.5	10.3	-3.1	9.3	-4.3	2.8	50	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	29.2	19.1	27.8	18.7	26.0	18.1	20.7	26.6	19.7	25.4	18.9	24.4	3.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.0	23.4	17.9	13.2	22.3	17.0	12.4	21.5	60.5	26.8	57.4	25.3	54.6	24.4	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.2	7.3	-26.0	31.5	3.5	2.1	-28.5	33.0	-30.6	34.2	-32.6	35.4	-35.1	36.9		
			-26.2	22.3	3.5	1.1	-28.7	23.1	-30.7	23.8	-32.7	24.4	-35.2	25.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-7.5	-7.0	-1.7	6.5	13.0	16.7	19.4	17.4	11.8	5.4	-1.0	-5.3
	DBStd	10.72	6.20	6.41	4.63	4.92	4.89	3.95	3.51	3.80	4.01	4.30	5.13	6.13
	HDD10.0	2520	542	477	364	128	26	2	0	0	26	153	329	474
	HDD18.3	4743	800	710	623	355	174	78	30	63	198	402	579	732
	CDD10.0	954	0	0	0	23	121	202	290	230	80	9	0	0
	CDD18.3	136	0	0	0	0	10	28	62	34	2	0	0	0
	CDH23.3	1129	0	0	0	2	91	241	488	292	16	0	0	0
	CDH26.7	286	0	0	0	0	15	51	137	83	1	0	0	0
(14) Wind	WSAvg	3.7	3.9	4.0	4.0	3.7	3.6	3.5	3.2	3.3	3.3	3.9	3.9	4.1
(15) Precipitation	PrecAvg	665	47	39	36	34	55	71	79	74	66	66	52	45
	PrecMax	878	84	78	73	57	116	118	167	145	185	134	87	80
	PrecMin	437	16	8	14	12	17	9	13	13	12	16	3	14
	PrecStd	108	17	14	16	13	25	26	40	37	33	36	19	19
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	3.7	12.9	23.0	28.2	30.1	33.0	32.8	25.2	18.9	11.1	7.1
		MCWB	3.1	2.4	7.1	13.4	18.1	19.1	20.3	19.3	17.2	12.7	9.3	6.6
	2%	DB	2.0	2.2	8.9	20.0	26.0	28.1	30.0	28.8	22.9	15.8	8.2	4.9
		MCWB	1.5	1.3	4.3	11.7	16.7	18.7	19.3	19.0	15.7	11.9	7.4	4.3
	5%	DB	1.1	1.8	6.1	17.1	23.2	26.1	28.0	26.2	20.2	13.1	6.9	2.9
		MCWB	0.7	1.0	2.9	10.3	15.1	17.8	18.9	18.4	14.9	10.6	6.1	2.4
	10%	DB	0.1	1.0	4.1	14.9	21.2	24.1	26.0	24.2	18.1	11.2	5.1	1.2
		MCWB	-0.2	0.4	2.0	9.0	13.9	16.7	18.6	17.7	13.7	9.2	4.4	0.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	2.7	7.2	14.3	19.4	21.5	22.5	21.6	18.5	14.1	9.8	6.6
		MCDB	3.3	3.1	11.7	21.6	26.7	27.5	28.3	27.2	23.1	16.8	11.2	6.9
	2%	WB	1.4	1.6	4.9	12.4	17.8	19.8	21.0	20.1	16.6	12.5	7.5	4.0
		MCDB	1.6	1.9	7.6	18.1	24.2	25.9	26.7	25.8	21.1	15.0	8.0	4.4
	5%	WB	0.5	0.8	3.3	11.0	16.4	18.7	20.0	19.2	15.4	11.0	5.9	2.2
		MCDB	0.8	1.2	5.2	16.2	21.9	24.3	25.4	25.0	19.3	13.3	6.4	2.5
	10%	WB	-0.3	0.2	2.2	9.7	15.0	17.5	19.2	18.2	14.3	9.7	4.3	0.7
		MCDB	-0.1	0.6	3.8	14.0	19.5	22.6	24.5	23.3	17.8	11.0	4.8	1.0
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.8	7.0	8.9	10.0	9.6	9.5	9.4	8.0	5.7	4.1	4.2
		MCDBR	3.7	4.5	9.7	12.4	12.4	12.4	12.1	12.1	11.3	8.8	5.6	3.7
	5% WB	MCWBR	3.7	4.1	6.4	6.7	5.9	5.5	4.6	5.2	5.9	5.9	5.0	3.6
		MCDWR	3.6	4.0	7.8	11.2	11.4	11.1	10.3	10.6	10.0	7.9	4.9	3.6
(40) Clear-Sky Solar Irradiance	taub	taub	0.291	0.330	0.347	0.423	0.402	0.388	0.435	0.440	0.390	0.357	0.322	0.287
		taud	2.120	2.058	2.116	2.108	2.289	2.351	2.243	2.209	2.328	2.356	2.236	2.120
	Edn at Noon	Edn at Noon	598	707	803	790	839	856	803	768	760	692	563	503
		Edh at Noon	62	97	116	137	121	115	126	123	96	73	56	49
(44) All-Sky Solar Radiation	RadAvg	0.60	1.45	2.99	4.00	5.25	5.49	5.37	4.36	2.79	1.27	0.49	0.32	
	RadStd	0.12	0.19	0.30	0.34	0.49	0.53	0.58	0.47	0.36	0.22	0.09	0.09	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(47) Regional (10 neighbors)	DBAvg	+0.78	N/A	N/A	N/A	N/A	-165	-228	+96
	DBAvg	+0.74	N/A	N/A	N/A	N/A	-158	-228	+113

Nomenclature: See separate page